Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Schrödinger equation is equivalent to information conservation

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy.lean`
- Library: `UnitaryEntropy`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, the Wigner library files `Density.lean` and `Differentiable.lean` (which imports the rest of `ForPhyslib/QuantumMechanics/Wigner`), and two new `ForMathlib` files:
  `ForMathlib/Analysis/Calculus/Deriv/Star.lean` (in a normed star algebra, the derivative of a path of unitaries through 1 is skew-adjoint, and $U^\dagger U$ is constant when $U' = XU$ with X skew-adjoint),
  `ForMathlib/Analysis/InnerProductSpace/Adjoint.lean` (in finite dimension, $U^\dagger U = 1$ already makes U unitary)
- Wigner library, in `ForPhyslib/QuantumMechanics/Wigner/`: `Density` (density operators, pure states `proj`, equal mixtures `mix`; moved there from this brief), `Basic`, `Normalize`, `Rigidity`, `Theorem` (Wigner's theorem without bijectivity, after Gehér, now in any

dimension, see Section 1.1; uniqueness up to one phase in finite dimension), `Projection` (the same for maps on rank-one projections), `Continuity` (Bargmann's triple product excludes the anti-unitary case near t_0), `Differentiable` (explicit, differentiable choice of the phase). Its `ForMathlib` helpers are `Analysis/Complex/ThreeCircles.lean`, `Analysis/InnerProductSpace/PhaseUnitary.lean` and `Analysis/InnerProductSpace/LinearMap.lean` (conjugate-linear isometries conjugate the inner product; polarization of $|x\rangle\langle y|$). Coordinate conjugation in an orthonormal basis came from TauCeti (`TauCeti.Analysis.InnerProductSpace.Conjugation`) until the library moved to Hilbert bases.

Setting. E is a finite-dimensional complex Hilbert space, and operators are continuous linear maps $E \to E$. A density operator (`IsDensity`) is a positive operator of trace 1. The pure state of a unit vector ψ is `proj` $\psi = |\psi\rangle\langle\psi|$ (Mathlib's `rankOne`), and `mix` $\rho\,\sigma = \frac{1}{2}\rho + \frac{1}{2}\sigma$. The von Neumann entropy `vonNeumann` is $\sum -\mu \log \mu$ over the roots μ of the characteristic polynomial. For self-adjoint ρ these are its eigenvalues (`vonNeumann_eq_sum`), and the definition makes invariance under conjugation immediate. `born` $\psi\,\phi = |\langle\psi|\phi\rangle|^2/(\|\psi\|^2\|\phi\|^2)$, and `mixEntropy` $p = H_2((1+\sqrt{p})/2)$ with H_2 Mathlib's `binEntropy`. The Schrödinger equation for a propagator is $U'(t) = (i\hbar)^{-1}H(t)\,U(t)$ with $H(t)$ self-adjoint.

Results.

Brief	Lean
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumann_mix_proj`, `mixEntropy_eq`
Corollary 1, strictly decreasing	`mixEntropy_strictAntiOn` (on $[0,1]$)
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \ldots$	`born_eq_iff` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary propagator	`star_mul_self_of_schrodinger`, `ContinuousLinearMap.mem_unitary_of_star_mul_self`
Theorem 2 ($\Rightarrow$), existence for constant H	`hasDerivAt_exp_schrodinger`: $U(t) = \exp(t(i\hbar)^{-1}H)$
Theorem 2 ($\Rightarrow$), DR-INFO	`vonNeumann_unitary_conj`, `IsDensity.unitary_conj`, `schrodinger_conserves_entropy`
Theorem 2 ($\Leftarrow$), Step 1	`vonNeumann_eq_zero_iff`: $S(\rho) = 0$ iff ρ is pure
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves`
Theorem 2 ($\Leftarrow$), Step 2, Wigner's theorem	`Wigner.wigner_rankOne` (from `Wigner.wigner`)
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`Wigner.eventually_exists_unitary`, `PhaseEqOnSphere.false_of_unitary_of_antiunitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)$	`Wigner.exists_unitary_differentiableAt`
Theorem 2 ($\Leftarrow$), Step 3	`exists_hamiltonian` (from `star_eq_neg_of_hasDerivAt_of_mem_unitary`), for $\hbar \neq 0$
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves`

Differences from the brief.

- This part is finite-dimensional; the brief has a separable Hilbert space. The calculus results (`star_mul_self_of_schrodinger`, `exists_hamiltonian`) hold in any Hilbert space for bounded $H(t)$. Corollary 1 and Theorem 2 through Wigner's theorem are redone in any dimension in Section 1.1.

- Corollary 1 is proved without diagonalizing the mixture. $\rho = \frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$ satisfies $\rho^3 = \rho^2 - \frac{1-\lambda}{4}\rho$ (`mix_proj_cubic`), so every eigenvalue is 0 or $(1 \pm \sqrt{\lambda})/2$. On these three points $-\mu \log \mu = \alpha\mu + \beta\mu^2$ for suitable α, β, and $\operatorname{tr}\rho = 1$, $\operatorname{tr}\rho^2 = (1+\lambda)/2$ fix the sum. This also covers the degenerate cases $\lambda = 0$ (double eigenvalue $\frac{1}{2}$) and $\lambda = 1$ (ρ pure).

B EVMIX is used only for equal mixtures, which is all Step 2 needs. For the ($\Rightarrow$) direction, $\rho \mapsto U\rho U^\dagger$ is linear, so it preserves all mixtures.

- DR-INFO is $S(\rho_I) = S(\rho_F) = S(\rho_P)$. The reconstructed past $U^\dagger \rho_F U$ is ρ_I itself, which is also proved.

- The ($\Rightarrow$) direction shows that every propagator solving the equation is unitary. Existence is shown for constant H only; for time-dependent $H(t)$ it is the linear ODE existence theorem, which the brief also assumes.

- `schrodinger_of_preserves` assumes $\Phi_{t_0} = \mathrm{id}$ and differentiability of $t \mapsto \Phi_t(\rho)$ at t_0 only on density operators, one ρ at a time. It concludes that $\Phi_t(|\psi\rangle\langle\psi|) = |U(t)\psi\rangle\langle U(t)\psi|$ for t near t_0, with $U(t)$ unitary for every t, $U(t_0) = 1$ and $U'(t_0) = (i\hbar)^{-1} H U(t_0)$ for a self-adjoint H. This is the Schrödinger equation at t_0, which is what EVDIFF at t_0 gives; the brief's two-time evolution maps give it at every initial time.

- Wigner's theorem is used in its non-bijective form (Gehér), so the bijectivity of the correspondence of pure states via the reconstructed past (Step 1) is not needed.

- Not formalized: $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for mixed ρ. It would need Φ_t to be affine (Kadison), while the brief assumes only equal mixtures, and the conclusion of the brief concerns U only.

Problems found in the brief. No errors. Implicit points:

- The "iff" in Corollary 1 needs $\lambda \in [0, 1]$, where the entropy is strictly decreasing in λ; for other λ, $\sqrt{\lambda}$ and the logarithms are not meaningful.

- Step 3 assumes a norm-differentiable unitary family, which forces a bounded generator. On an infinite-dimensional Hilbert space it therefore covers bounded Hamiltonians only. The brief "Multiple equivalent characterizations of the Schrödinger equation" treats the general case with Stone's theorem.

- In Step 3, $\hbar \neq 0$ is needed to write $A = H/(i\hbar)$.

- Wigner's theorem gives each $U(t)$ only up to a phase, and Step 3 differentiates $U(t)$. A differentiable choice exists. Extend Φ_t to all $|x\rangle\langle y|$ by polarization, so that it sends $|x\rangle\langle y|$ to $|Ux\rangle\langle Uy|$, and fix an orthonormal basis (b_k) and an index i. Then

$$U(t) = \left(\operatorname{Re}\langle b_i|\Phi_t(|b_i\rangle\langle b_i|)\, b_i\rangle\right)^{-1/2} \sum_k \Phi_t(|b_k\rangle\langle b_i|)\ |b_i\rangle\langle b_k|$$

  is the unitary with $\langle b_i|U(t)b_i\rangle > 0$. It is a finite expression in values of Φ_t on pure states, so it is differentiable at t_0 whenever these are.

- "Joint continuity and $\Phi_{t_0} = I$ exclude the anti-unitary case" needs only continuity at t_0 of $t \mapsto \Phi_t(P)$ for three pure states P: the imaginary part of Bargmann's triple product $\mathrm{tr}(\Phi_t(P_1)\Phi_t(P_2)\Phi_t(P_3))$ is $\frac{1}{4}$ at t_0 and would be $-\frac{1}{4}$ for an anti-unitary. In dimension 1 there is nothing to exclude.

1.1 Hilbert spaces of any dimension

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/InfiniteDimensional.lean`
- Library: `UnitaryEntropy`, which now builds every module `UnitaryEntropy.*`
- Imports:
  `ForPhyslib.QuantumMechanics.VonNeumannEntropy`,
  `ForMathlib.Analysis.InnerProductSpace.StrongDerivative`,
  `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison` (eigenvector action of the functional calculus, for the comparison in finite dimension),
  and the finite-dimensional part `UnitaryEntropy` (for `born`, `mixEntropy`, `mix_proj_cubic`, `mix_proj_sq`, `star_mul_self_of_schrodinger` and `exists_hamiltonian`, none of which assumes finite dimension)
- New file `ForPhyslib/QuantumMechanics/DensityOperator.lean`: density operators on a Hilbert space of any dimension, on top of the trace-class operators of PhyslibAlpha (`PhyslibAlpha/ProbabilisticTheory/HilbertSpace/TraceClass/`, from the physlib dependency; it builds against our Mathlib pin). Pure states, convex combinations, conjugation by an isometry, idempotent density operators are pure, and agreement with `IsDensity` in finite dimension.
- New file `ForPhyslib/QuantumMechanics/VonNeumannEntropy.lean`: the von Neumann entropy in $[0, \infty]$, its invariance under conjugation by an isometry, and $S(\rho) = 0$ iff ρ is pure.
- New `ForMathlib` material:
  `ForMathlib/Analysis/InnerProductSpace/PositiveTrace.lean` (the trace of a positive operator in $[0, \infty]$, independent of the Hilbert basis),
  `ForMathlib/Analysis/InnerProductSpace/HilbertBasisMaps.lean` (for a Hilbert basis: a vector is determined by its coordinates, Parseval for their moduli, the diagonal unitary $b_i \mapsto \delta_i b_i$, coordinatewise conjugation, the isometry $b_i \mapsto v_i$ onto an orthonormal family, and $V(V^\dagger x) = x$ when $\|V^\dagger x\| = \|x\|$),
  `ForMathlib/Analysis/InnerProductSpace/StrongDerivative.lean`
  (by Banach–Steinhaus, the strong derivative of an operator family is a bounded operator; for unitaries with $U(t_0) = 1$ it is skew-adjoint).
- The Wigner library `ForPhyslib/QuantumMechanics/Wigner/` (`Basic`, `Normalize`, `Rigidity`, `Theorem`) now works over a Hilbert basis of any Hilbert space. `Theorem` proves Wigner's theorem in any dimension, with isometries, and derives the finite-dimensional theorem from it. `Continuity` excludes the anti-unitary branch in any dimension, without a trace, and derives the finite-dimensional exclusion from it. `Differentiable` adds the differentiable choice of the phase in any dimension. The other files built on the library are unchanged.

Setting. E is a complex Hilbert space of any dimension; separability is not used. `posTrace` $T = \sum_i \langle e_i|Te_i\rangle \in [0,\infty]$ for a Hilbert basis (e_i). For positive T it does not depend on the basis (`posTrace_eq_tsum`): write $\langle e_i|RR^\dagger e_i\rangle = \|R^\dagger e_i\|^2$, expand by Parseval in a second basis and exchange the two sums of nonnegative terms. A density operator (`IsDensityOperator`) is a positive operator that is trace class in the sense of PhyslibAlpha (`IsTraceClass`) with `trace` $\rho = 1$; equivalently, $0 \le \rho$ and `posTrace` $\rho = 1$ (`isDensityOperator_iff`). The entropy is `vonNeumannEntropy` $\rho =$ `posTrace` $\eta(\rho)$ with $\eta(x) = -x\log x$ and $\eta(\rho)$ given by the continuous functional calculus. The spectrum of a density operator lies in $[0,1]$, where $\eta \ge 0$, so $\eta(\rho)$ is positive. `proj`, `mix`, `born` and `mixEntropy` are as above.

Results.

Brief	Lean
QST:H, density operators	`IsDensityOperator`, `isDensityOperator_proj`, `IsDensityOperator.mix`
QST:H, von Neumann entropy	`vonNeumannEntropy`
Agreement in finite dimension	`isDensityOperator_iff_isDensity`, `vonNeumannEntropy_eq_ofReal_vonNeumann`
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumannEntropy_mix_proj`
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \dots$	`born_eq_iff_vonNeumannEntropy` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary conjugation preserves the entropy	`IsDensityOperator.vonNeumannEntropy_conj` (for isometries)
Theorem 2 ($\Rightarrow$), DR-INFO	`IsDensityOperator.conj`, `schrodinger_conserves_vonNeumannEntropy`
Theorem 2 ($\Leftarrow$), Step 1	`IsDensityOperator.vonNeumannEntropy_eq_zero_iff`
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves_vonNeumannEntropy`, `exists_isWignerMap_of_preserves`
Wigner's theorem, any dimension	`Wigner.IsWignerMap.exists_linearIsometry_or_conjLinearIsometry`, `Wigner.wigner_isometry`
Wigner's theorem, onto the rays	`Wigner.IsWignerMap.exists_unitary_or_antiunitary_of_surjective`
Theorem 2 ($\Leftarrow$), Steps 1–2 with Wigner	`exists_isometry_of_preserves`; `exists_unitary_of_preserves` if every pure state is an image
Theorem 2 ($\Leftarrow$), surjectivity from the reconstructed past	`exists_proj_of_past`, `exists_unitary_of_preserves_of_past`
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`eventually_exists_unitary_of_preserves` (from `Wigner.eventually_exists_unitary_of_unitary_or_antiunitary`)
Theorem 2 ($\Leftarrow$), surjectivity is needed	`shift_preserves_not_unitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)\psi$	`Wigner.exists_unitary_differentiableAt_apply`
Theorem 2 ($\Leftarrow$), Step 3	`ContinuousLinearMap.exists_hasDerivAt_apply`, `star_eq_neg_of_hasDerivAt_apply_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves_vonNeumannEntropy`

Differences from the brief.

- Wigner's theorem in any dimension follows Gehér's proof with a Hilbert basis (b_k). The first normalization changes: the images $f(b_k)$ are orthonormal but need not span, so they define an isometry V with $Vb_k = f(b_k)$, not a unitary. By Parseval, $\|V^\dagger f(v)\|^2 = \sum_k |\langle f(b_k)|f(v)\rangle|^2 = \sum_k |\langle b_k|v\rangle|^2 = 1 = \|f(v)\|^2$, so $f(v)$ lies in the range of V and $V^\dagger \circ f$ is a Wigner map that fixes every b_k. The diagonal unitary and the conjugation are then built from the basis as in finite dimension. The result is a linear or conjugate-linear isometry; it is onto, hence unitary or anti-unitary, when every ray is the image of a ray, and always in finite dimension.

- The anti-unitary branch is excluded by continuity without a trace. For $P = |a\rangle\langle a|$, Q, R rank-one projections onto a, b, c and a fixed vector x, $\langle x|PQRP\,x\rangle = |\langle a|x\rangle|^2\,\Delta(a,b,c)$ (`inner_rankOne_mul_rankOne_mul_rankOne_mul_rankOne_apply`). Take the test triple $b_i, (b_i - b_k)/\sqrt{2}, (b_i - ib_k)/\sqrt{2}$ with $\Delta = (1+i)/4$, $x = b_i$, and the images $\Phi_t(P)$. The imaginary part is $\frac{1}{4}$ at t_0 and continuous at t_0 in operator norm, but ≤ 0 for an anti-unitary, since its triple product is $\overline{\Delta}$ and the factor $|\langle a|x\rangle|^2$ is nonnegative. The finite-dimensional `Wigner.eventually_exists_unitary`, which used $\mathrm{tr}(PQR) = \Delta$, is now a corollary. Continuity is assumed in operator norm; continuity in trace norm implies it.

- EVDIFF is formalized as in finite dimension: $t \mapsto \Phi_t(\rho)$ is differentiable at t_0 in operator norm for every density operator ρ. The conclusion is the Schrödinger equation on vectors, as in DR-SCEQ: $\psi(t) = U(t)\psi$ satisfies $\psi'(t_0) = (i\hbar)^{-1}H\psi(t_0)$ for every ψ, with one bounded self-adjoint H. It is not the equation for $t \mapsto U(t)$ in operator norm, which the finite-dimensional `schrodinger_of_preserves` gives. The basis sum that fixes the phase in finite dimension does not converge in operator norm; instead, for a fixed unit vector e, Φ_t extended by polarization gives $(|Ux\rangle\langle Ue|)\,e = \langle Ue|e\rangle\,Ux$, and dividing by $|\langle Ue|e\rangle|$ fixes the phase one vector at a time. So $t \mapsto U(t)x$ is differentiable at t_0 for each x. By Banach–Steinhaus, applied to the difference quotients, the derivatives $x \mapsto (U(\cdot)x)'(t_0)$ form a bounded operator A. Differentiating $\langle U(t)x|U(t)y\rangle = \langle x|y\rangle$ gives $A^\dagger = -A$, and $H = i\hbar A$.

- No spectral decomposition is used. The entropy is the trace of $\eta(\rho)$, and pure states are found through the spectrum: $S(\rho) = 0$ forces $\eta(\rho) = 0$, so η vanishes on the spectrum, which then lies in $\{0,1\}$; ρ is idempotent, and an idempotent density operator is $|\psi\rangle\langle\psi|$ (`IsDensityOperator.eq_proj_of_isIdempotentElem`).

- Corollary 1 uses the cubic relation as in finite dimension, with the spectral mapping theorem for polynomials in place of eigenvalues. Then $\eta(\rho) = \alpha\rho + \beta\rho^2$, and its trace is computed from $\sum_i \langle e_i|x\rangle\langle y|e_i\rangle = \langle y|x\rangle$ for the rank-one terms.

- The $(\Rightarrow)$ direction uses only $U(t)^\dagger U(t) = 1$, which every propagator satisfies. Conjugation $X \mapsto UXU^\dagger$ by an isometry is a non-unital star algebra homomorphism, so it commutes with the functional calculus of η (since $\eta(0) = 0$), and it preserves `posTrace`. Unitarity is not needed.

- Entropies are compared in $[0, \infty]$; in infinite dimension $S(\rho) = \infty$ occurs.

- DR-INFO is formalized in full: $S(\Phi\rho) = S(\rho)$, and every density operator ρ has a reconstructed past, a density operator σ with $\Phi\sigma = \rho$; then $S(\sigma) = S(\rho)$ by the first condition. The finite-dimensional part does not need the past, and `exists_isometry_of_preserves` shows what remains without it.

Problems found in the brief.

- In infinite dimension the reconstructed past in DR-INFO is essential, and Step 1 uses it when it says that pure states "correspond bijectively". DR-INFO presupposes a past ensemble ρ_P for every initial ensemble, so Φ is onto the density operators, as in the introduction's "deterministic and reversible". Then every pure state is the image of a pure state, since its past has entropy zero (`exists_proj_of_past`), and Wigner's theorem gives a unitary or an anti-unitary (`exists_unitary_of_preserves_of_past`). The brief's proof is therefore correct, but the past cannot be dropped. Conjugation $\rho \mapsto S\rho S^\dagger$ by the shift S on $\ell^2(\mathbb{N})$ maps density operators to density operators and preserves the entropy and equal mixtures, but no pure state goes to $|e_0\rangle\langle e_0|$. So it is induced by no unitary and by no anti-unitary (`shift_preserves_not_unitary`), and Wigner's theorem gives only an isometry. In finite dimension the past is automatic.

- The time structure of [B]EVMIX does not replace the past. On $L^2(0,\infty)$ the right shifts $(V_t f)(x) = f(x-t)$ for $x > t$ and 0 otherwise are isometries forming a strongly continuous semigroup with $V_0 = I$. Then $\Phi_{t_0\to t_1}(\rho) = V_{t_1-t_0}\rho V_{t_1-t_0}^\dagger$ satisfies [B]EVMIX, preserves the entropy, and is induced by no unitary for $t_1 > t_0$; it has no reconstructed past. (By the Wold–Cooper decomposition, every strongly continuous semigroup of isometries is a unitary group plus copies of this one.) Not formalized. A norm-continuous semigroup of isometries is $V_t = e^{tA}$ with bounded A, and differentiating $V_t^\dagger V_t = I$ at 0 gives $A^\dagger = -A$, so it is unitary. The example therefore needs an unbounded generator, which Step 3 does not cover anyway.

- EVDIFF for every density operator forces a bounded Hamiltonian, as the Banach–Steinhaus step shows. Unbounded Hamiltonians, whose evolution is differentiable only on a dense set of states, are outside the theorem as stated; the brief "Multiple equivalent characterizations of the Schrödinger equation" treats them with Stone's theorem. Otherwise no problems beyond those of the finite-dimensional part.